

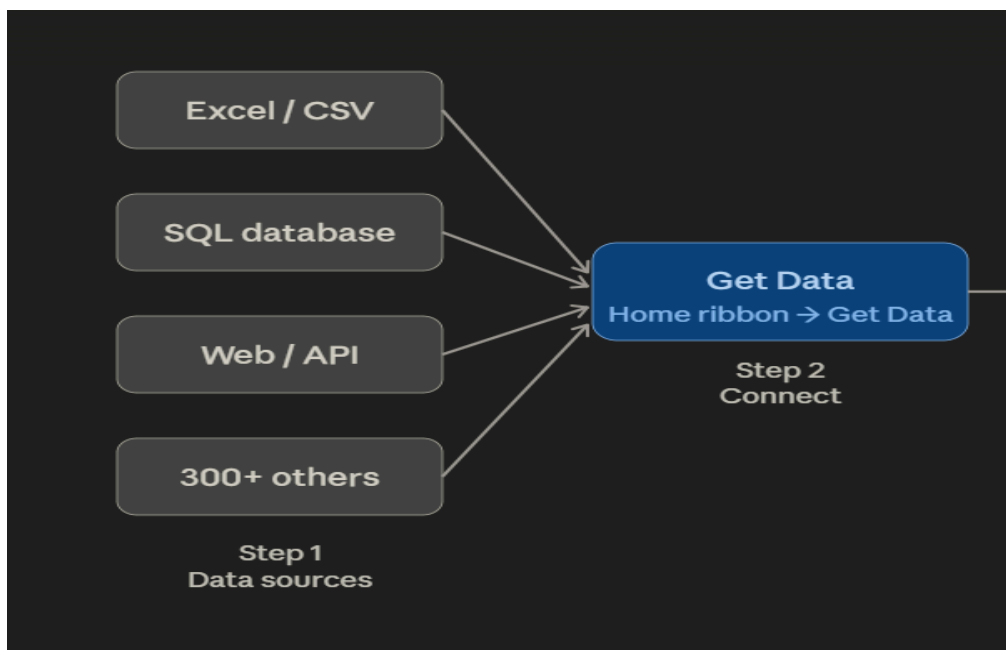
CONNECTING TO DATA

Concept: What is Data Connection?

- **Definition:** The process of linking Power BI to external data sources.
- **Purpose:** To provide Power BI with the **raw data** required for data modeling and visualization.
- **Capacity:** Power BI can connect to **300+ data sources** through a single "Get Data" button.

Common Data Source Categories

- **Files:** Excel, CSV, PDF, JSON, XML
- **Databases:** SQL Server, MySQL, PostgreSQL, Oracle
- **Cloud:** Azure, Google Analytics, Salesforce
- **Web:** Any URL/API endpoint
- **Others:** SharePoint, Power BI Datasets, Dataverse



Data Connection Methods

1. Connect to Excel / CSV (Most Common)

- Step 1. Click **Home** → **Get Data** → **Excel Workbook** (or **Text/CSV** for CSV files).
- Step 2. Browse to and **Open** your file.
- Step 3. The **Navigator** window appears, showing sheets/tables.
- Step 4. Check the box next to the required sheet/table.
- Step 5. Review the data **preview**.

Step 6. Select **Load** (as-is) OR **Transform Data** (to clean).

Note: Power BI automatically detects the delimiter (comma, tab, etc.).

2. Connect to SQL Server

Step 1. **Home** → **Get Data** → **SQL Server**.

Step 2. Enter **Server name** and optional **Database name**.

Step 3. Choose data connectivity mode: **Import** or **DirectQuery**.

Step 4. Enter credentials (if prompted).

Step 5. Select tables in the Navigator → Click **Transform Data**.

Data Connectivity Modes for Data Bases: Import vs DirectQuery

Mode	What It Does	When to Use
Import	Copies data into Power BI's internal model.	Most reports, as it offers faster performance.
DirectQuery	Queries the database live every time a report is viewed.	When data changes every second (real-time needs).

Example: Superstore Sales Dataset

This dataset is used for hands-on practice in all future modules (data modeling, DAX, dashboards).

Table	Content
Orders	Order ID, Date, Customer, Product, Sales, Profit, Quantity
Returns	Which orders were returned
People	Sales rep assigned to each region

Load the Superstore Dataset

Step 1. Download the Superstore Excel file.

Step 2. Open Power BI Desktop.

Step 3. **Home** → **Get Data** → **Excel Workbook**.

Step 4. In the Navigator, check **Orders**, **Returns**, and **People**.

Step 5. Click **Transform Data** (not Load).

Step 6. Power Query Editor opens (with 3 queries).

Step 7. Click **Close & Apply**.

Step 8. Switch to **Table View** to verify the loaded data.

DIFFERENCE BETWEEN LOAD AND TRANSFORM

"Load" imports data directly without changes.

"Transform Data" is preferred as it opens Power Query to clean and reshape data before it enters the model.

Feature	Load	Transform Data
Data accessed?	Yes	Yes
Power Query opens?	No — skipped	Yes — you review first
Data enters model?	Immediately	Only after you click Close & Apply
Can you fix issues?	Only after loading (messy)	Before loading (clean)

Why Transform?

Addressing data issues early prevents broken visuals and calculation errors later.

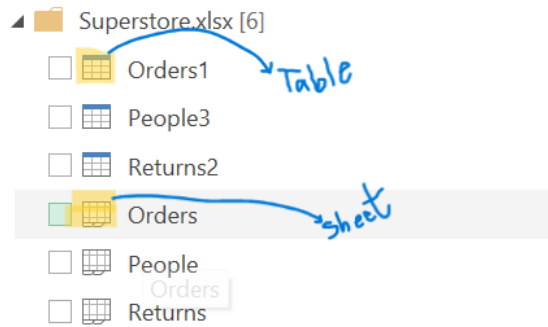
Common Errors and Fixes

Error	Cause	Fix
File not found on refresh	Excel file moved or renamed	Right-click query → Data Source Settings → Update path
Column headers show as Column1, Column2	Headers are in Row 1, not recognised	Transform Data → Use First Row as Headers
Date column shows ABC icon	Date imported as text	Click column type icon → Change to Date
Blank rows at top	Junk rows above headers	Remove Top Rows in Power Query

NOTE

It is suggested to select tables instead of sheets in the navigator

- How do we differentiate : based on the logo



- If tables are not present
- Go to the Excel sheet → select the entire data → Ctrl +T